RIG-CLOSED PRIMES OVER A JACOBSON QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-rig-closed-jacobson; the proof environment immediately following it is its own proof

Editorial note. The label lemma-rig-closed-jacobson names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- rig-closed primes of a WAdm^Noeth arrow and the relative residue field characterization (lemma-rig-closed-point-relative-residue-field, lemma-rig-closed-point-relative of the same chapter)
- finite type arrows stay finite type after reduction (lemma-finite-type-finite-type-red)
- Jacobson rings: a finite type algebra over a Jacobson ring is Jacobson and its primes contract to maximal-mod-primes; in particular a field of finite type over a Jacobson ring is finite over it (algebra-lemma-silly-jacobson)

Source reference. Original lemma 0CY7: sources/stacks-chow/tags.txt, source lines 16232–16232. Prerequisite pointer invoked at proof line(s) 3396; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GC5: sources/stacks-restricted/pinned-original-source.tex, source lines 2738–2738. Prerequisite pointer invoked at proof line(s) 3392; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGB: sources/stacks-restricted/pinned-original-source.tex, source lines 3333–3333. Prerequisite pointer invoked at proof line(s) 3397; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGA: sources/stacks-restricted/pinned-original-source.tex, source lines 3314–3314. Prerequisite pointer invoked at proof line(s) 3391; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\varphi : A \to B$ be an arrow of $WAdm^{Noeth}$ which is adic and topologically of finite type. Let $\mathfrak{q} \subset B$ be rig-closed. If A/I is Jacobson for some ideal of definition $I \subset A$, then $\mathfrak{p} = \varphi^{-1}(\mathfrak{q}) \subset A$ is rig-closed.*

Proof. By Lemma 0GGA (combined with Lemma 0GC5) the residue field κ of $B/\mathfrak{q}$ is of finite type over $A/\mathfrak{a}$. Since $A/\mathfrak{a}$ is Jacobson, we see that κ is finite over $A/\mathfrak{a}$ by Algebra, Lemma 0CY7. We conclude by Lemma 0GGB. □